Sathya Devarajan

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Education

Purdue University — John Martinson Honors College West Lafayette, IN
B.S. Mechanical Engineering, GPA: 3.8/4.0 *Expected May 2028*

Experience

Lockheed Martin Sikorsky Stratford, CT
Tooling Design Engineer Intern *June 2026 – August 2026*

- Designed and revised 14 production tooling fixtures and jigs (drill jigs, assembly locators, masking fixtures) in CATIA V5 for CH-53K, Black Hawk, and search-and-rescue rotorcraft, 3 of which are now in use on the shop floor
- Toleranced 12 tools to ASME Y14.5 GD&T, matching every callout to the process that would actually make the part (CNC milling, manual machining, or additive) rather than to a single default standard
- Traced constraint and leakage failures in 3 released production tools back to root cause and redesigned each one, which cut procurement lead time and dropped a secondary machining setup from the build
- Maintained multi-level BOMs for all 14 tools & routed engineering change orders through SAP ERP & 3DEXPERIENCE PLM, so manufacturing engineers & machinists worked from current tooling data across releases
- Co-designed the hybrid-electric powertrain for the dual-propeller Nomad VTOL, packaging an ICE, motor generator, battery pack, and a per-propeller motor and planetary gearbox inside thermal limits; presented to leadership

Purdue Electric Racing (FSAE Electric) West Lafayette, IN
Drivetrain Engineer *August 2025 – Present*

- Designed the rear upright & planet carrier for the 2027 outrunner drivetrain in NX, cutting weight **32%**
- Ran static structural FEA on the rear upright & hub group, then reworked stiffener & fillet geometry against the stress peaks and 2 mm deflections it exposed, and pulled dead weight out of near-zero-stress areas
- Drafted the rear upright to sub-0.003 in. bearing & seal bore tolerances, called out for a manual boring finish
- Cut wheel-upright machining time 50% by reworking the toolpaths from 3-axis to 5-axis CNC milling, then built the fixture plate that locates and holds the upright through the remaining 3-axis production runs

Projects

Custom Electric Bike Design October 2025 – Present

- Built a fully parametric frame assembly in Siemens NX keyed to off-the-shelf tubing dimensions, so wheelbase, head angle, chainstay angle, and dropout geometry all update from a single set of driving inputs
- Ran Ansys FEA on each candidate frame, swapping stock tube diameters and wall thicknesses through the parametric model to find the lightest combination that holds a 2.0 minimum safety factor under static load
- Ran cost-versus-performance trade studies on motors, sprockets, and tubing alongside the FEA, converging on the cheapest catalog set that still meets the target safety factor and ride geometry, 9% under the first build cost

Autonomous Supply Robot March 2026 – April 2026

- Navigated a walled maze on ultrasonic and IMU sensing, adding infrared to dodge heat and magnetic hazards
- Designed a gripping mechanism that holds cargo in transit and releases it at the target destination
- Logged the path traveled by IMU dead-reckoning, recording hazard and delivery coordinates into a matrix

Complex Surface Topography December 2025 – January 2026

- Produced a 2D grayscale gradient heightmap from a source portrait photo
- Built 3D mesh geometry from the heightmap in Fusion 360, keeping the organic topology machinable
- Cut machining time 60% by reworking CAM toolpaths, holding surface detail with almost no hand rework

Mars Rover October 2025 – December 2025

- Built a compact drivetrain with a custom worm-gear cargo system to carry 500 g loads up 35° inclines
- Wrote a Python control loop that trims motor PWM from live IMU feedback to hold the line over rough terrain

Ethanol Production Process Optimization October 2025

- Built a low-fidelity Python model of an ethanol plant, parameterized by machinery, piping, and valve specs
- Swept every valid component combination to find the plant configuration with the lowest cost per unit energy

Skills

CAD/CAM/PLM: CATIA V5, Siemens NX, Fusion 360, 3DEXPERIENCE, Teamcenter, SAP ERP
Manufacturing: 3- & 5-Axis CNC Machining, Manual Machining, Additive Manufacturing, Fixture Design, Soldering
Tolerancing & Analysis: GD&T (ASME Y14.5), Ansys FEA, Dive CFD
Modeling: Python trade studies, combinatorial optimization, cost & energy modeling
Controls/Electronics: Python, MATLAB, Arduino, LTSpice, KiCad